WBN HAUL PLAN

How we turn trucks into trips and tonnes

A short walkthrough of the logic. No hidden factors. Built for review.

1

Time one trip

Road + load/dump + extras

2

Count trips

1,440 min / that time

3

Tonnes

trips x payload x trucks

4

Share the road

Same section, same delay

For colleagues - confirm the logic, then we keep it.

THE ONLY TRIPS FORMULA

How many trips one truck makes in a day

trips per truck per day

$$= \frac{\text{1,440 minutes (two 12-hour shifts)}}{\text{time of one trip} + \text{overhead}}$$

1,440 minutes

(two 12-hour shifts)

time of one trip + overhead

driving, loading, dumping — breaks, dispatch, shift change

EXAMPLE (illustration, not a named route)

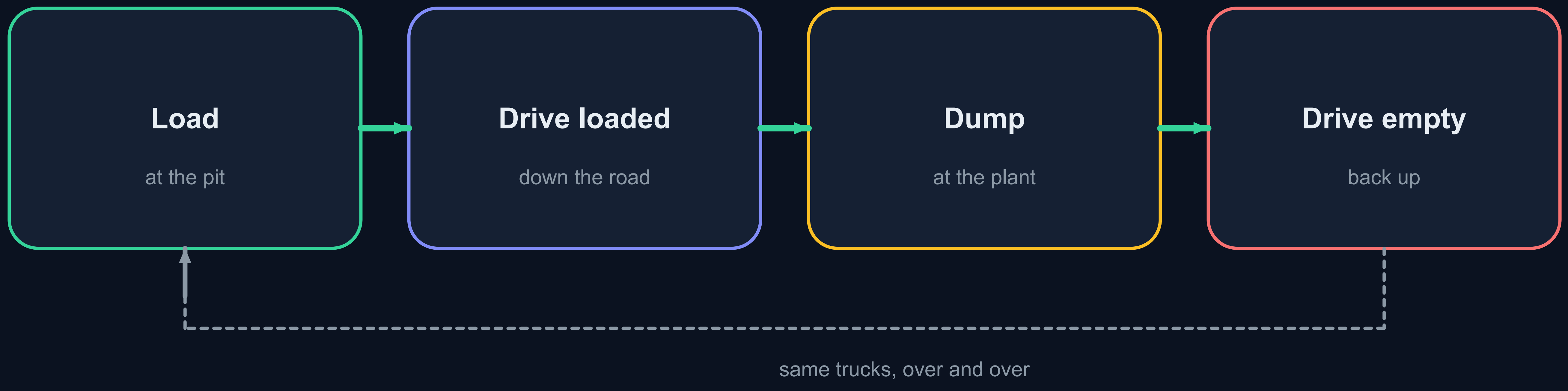
Trip 400 min + overhead 220 min = 620 min

1,440 / 620 = 2.3 trips/truck/day

60 trucks x 2.3 x 48 t = 6,624 t that day

ONE TRIP IS A LOOP

The truck comes back to the same loader



Trip time we use

Free road
empty-road time from history

+ Road extra
when the section is busy

+ Ops
load, spot, dump

+ Wait
loader queue (Congestion)

+ Bunching
trucks arriving in clumps

ROAD DELAY

Busy road, extra minutes. Empty road, none.

The extra time

15% x (how full)^4 x free time

Quarter full: almost nothing. Half full: still small. Nearly full: delay grows fast. That is the standard road curve.

How full is the section?

trucks per hour / capacity

Capacity = slowest posted speed / 50 m gap, one loaded lane. Empty return uses the other carriageway.

S1 TF - KR

600 / hr

one loaded lane

S2 KR - POS 12

600 / hr

one loaded lane

S3 POS 12 - KM 15

600 / hr

one loaded lane

S4 KM 15 - coast

400 / hr

one loaded lane

THE OTHER MINUTES

Three extras. Only one of them is the road.

Loader wait

Same trucks keep coming back to the same faces. More loaders, shorter wait. We never also add this wait onto the road.

Bunching

Trucks do not arrive perfectly spaced. Clumps add a little time. Capped at 3x the typical-fleet amount.

Overhead

Breaks, dispatch, shift change, fuel. Fitted so at a typical fleet the model matches real dispatch. Attached to each trip, not the clock.

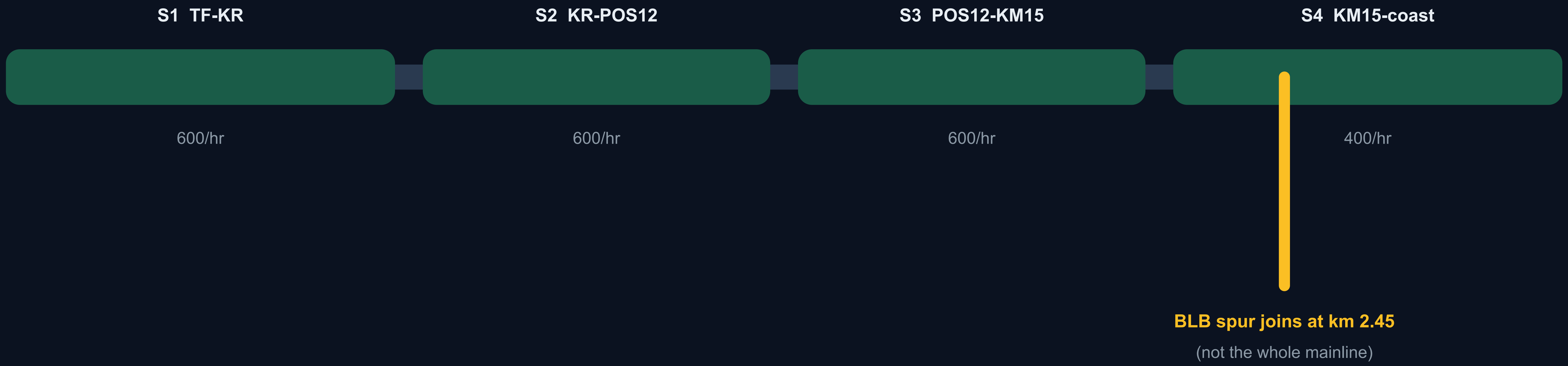
Where it is used

Plan / Allocate = road + ops + overhead.

Congestion tab also adds loader wait and bunching, so you can see them.

ONE ROAD, FOUR SECTIONS

If you share a section, you share its delay



Our trucks

Every haul on a section adds to that section's flow. TF to HUAFEI and TF to POS share S1.

Other companies

They count on the road. They do not count in our tonnes.

TWO NUMBERS. NEVER ONE BLEND.

Prediction and Achievable are different clocks

PREDICTION

Cycle model

Trips from the formula on page 2, times payload, times trucks. This is Plan and Allocate.

tonnes = trips x payload x trucks

ACHIEVABLE

Ticket engine

What the weighbridge history says a fleet can move, with a loader clip. This is Run scenario.

We show both. We never average them.

Pictures of crowding, GPS particles, and rain on Simulate tonnes do not rewrite either clock.

ALLOCATE + HARD RULES

Fill targets in order. Do not invent trucks.

P1 SAP

to FENI first

→

P2 LIM-TOS

limonite to plant

→

P3 LIM-LD

leftover to dump

- Trucks keep their contractor. BLB is RIM only. KR is SMA only.
- IWIP POS trucks are their own fleet. Not taken from RIM or SMA.
- Road crowding is a picture of the plan. It does not cut tonnes.
- We do not apply 0.85 availability. Downtime is already inside trip time.